Types of Chemical Reactions

Classify, predict and balance

Name _____ Class _____ Date _____

ACTIVITY SERIES AND DATA

K > Na > Ca > Mg > Al > Zn > Fe > Pb > (H) > Cu > Ag > Au · halogens: F > Cl > Br > I

Molar masses (g/mol): Mg 24.31 · MgO 40.31 · Zn 65.38 · Cu 63.55

Balance every equation. Show units in every calculation step.

A. Classifying

1. Classify each reaction: (a) $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$ (b) $2\,HgO \longrightarrow 2\,Hg + O_2$ (c) $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2\,NaCl$ (d) $2\,Al + 3\,CuCl_2 \longrightarrow 2\,AlCl_3 + 3\,Cu$ (e) $C_2H_4 + 3\,O_2 \longrightarrow 2\,CO_2 + 2\,H_2O$

B. Predicting products

2. Predict whether a reaction occurs. If it does, write the balanced equation. (a) Mg + $ZnCl_2$ (b) Ag + $CuSO_4$ (c) Zn + HCl (d) Cl_2 + NaBr (e) Br_2 + NaCl

3. Complete and balance: (a) Na + Cl_2 → (b) $MgCO_3$ → (on heating) (c) H_2SO_4 + NaOH →

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C. Combustion

4. Write balanced equations for the complete combustion of (a) methane, CH_4 (b) butane, C_4H_{10}. (c) Name a harmful product of incomplete combustion.

D. Double displacement

5. Explain why mixing solutions of silver nitrate and sodium chloride gives a reaction, but mixing potassium nitrate and sodium chloride does not.

E. Calculations

6. 2.43 g of magnesium burns completely in oxygen: $2\,Mg + O_2 \longrightarrow 2\,MgO$. Calculate the mass of magnesium oxide formed.

7. Excess zinc is added to a copper(II) sulfate solution, and 4.86 g of copper is produced. Calculate the mass of zinc that reacted.

F. Challenge

8. A student places strips of metals X, Y and Z in solutions of each other's salts. X displaces Y and Z; Z displaces Y. Put X, Y and Z in order of reactivity, and explain.

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